

Exercise – *Harry Potter*

“What I don’t understand, though—just out of curiosity—I mean, would one Horcrux be much use? Can you only split your soul once? Wouldn’t it be better, make you stronger, to have your soul in more pieces, I mean, for instance, isn’t seven the most powerfully magical number, wouldn’t seven—?”

Tom Riddle to Prof. Slughorn, Harry Potter and the Half-Blood Prince by J. K. Rowling

Before he became Lord Voldemort, he was Tom Riddle. What knowledge about this bright young boy can unlock the key to the powerful wizard’s defeat? After pursuing this question for a long time, Headmaster Dumbledore has discovered that Professor Slughorn holds a crucial but, for some reason, secret memory about the Dark Lord’s past. Unable to extract it himself, Dumbledore believes that Harry Potter has what it takes to complete the task.

Harry’s first idea is to use the truth potion *Veritaserum* on Slughorn. However, since the professor is well-versed in the art of potion-making, planting the whole dose into his tea might be too risky. Instead, Harry plans to split the potion into arbitrary parts and let them circulate among his friends until the right moment when they can drop the bits into the professor’s next meal. The advantage of such a scheme would be that alternative routes for the potion might raise less suspicion. The disadvantage, given the highly volatile nature of the potion, would be that a total dose too small to have any effect might find its way to Slughorn.

Harry’s circle of friends consists of n people $p_1, p_2, p_3, \dots, p_n$. We assume that p_1 is Slughorn and p_2 is Harry himself, while the sequence $p_3, \dots, p_n$ is sorted decreasingly by the amount of trust Harry puts in the respective person. For some of the pairs (p_i, p_j) , parts of the potion may change hands from p_i to p_j . For every such pair, Harry has noted down two numbers d_{ij} and e_{ij} . Firstly, d_{ij} denotes the proportional amount of suspicion raised by such an exchange. Secondly, e_{ij} denotes its efficiency. That is, if p_i were to hand over x units of the potion, then the raised suspicion would be $d_{ij}x$ and p_j would only receive $e_{ij}x$ units of the potion.

Harry wonders if there is a way to deliver an effective dose to Slughorn without raising too much suspicion and by involving only his most trusted friends. More precisely, for numbers a, b, c , given an initial a units of the potion, is it possible for Slughorn to receive at least b units such that the sum of suspicions raised by all exchanges is at most c ? If yes, what is the smallest integer $k \geq 2$ such that involving only $p_1, \dots, p_k$ is sufficient and, for such k , what is the smallest possible integer s such that the suspicion raised by each pair (p_i, p_j) is at most s ?

Input The first line of the input contains the number $t \leq 30$ of test cases. Each of the t test cases is described as follows.

- It starts with a line containing five integers $n \ m \ a \ b \ c$, separated by a space. They denote
 - n , the number of people including Slughorn and Harry ($2 \leq n \leq 32$);
 - m , the number of pairs of people that may exchange the potion ($0 \leq m \leq 64$);
 - a , the amount of potion in Harry’s possession ($1 \leq a \leq 2^{24}$);
 - b , the amount of potion needed to have the desired effect on Slughorn ($1 \leq b \leq 2^{24}$);

- c , the maximum sum of suspicions that may be raised ($0 \leq c \leq 2^{24}$).
 - The following m lines define the different ways that the potion may be exchanged. Every line contains five integers $i \ j \ d \ e' \ e''$, separated by a space. They denote
 - i and j , the indices of the two respective persons p_i and p_j ($1 \leq i, j \leq n$);
 - $d_{ij} = d$, the raised suspicion as explained above ($0 \leq d \leq 2^{24}$);
 - $e_{ij} = \frac{e'}{e''}$, the efficiency as explained above ($0 < e' \leq e'' \leq 2^{24}$).
- You may assume $i \neq j$ and that any ordered pair (i, j) appears at most once in the input.

Output The output for each test case consists of a separate line. If the required amount of the potion cannot be handed over to Slughorn in a way that the sum of raised suspicions is at most c , then the output is “Busted!”. Otherwise, the output consists of two integers $k \ s$, separated by a space. The numbers k and s are defined as explained above.

Points There are four groups of test sets, worth 100 points in total.

1. For the first group of test sets, worth 30 points, you may assume that p_n must be involved. In other words, the correct output is either “Busted!” or $n \ s$. In addition, for the latter case you may assume that $s = 7777777$ (seven sevens).
2. For the second group of test sets, worth 30 points, you may assume that p_n must be involved.
3. For the third group of test sets, worth 20 points, you may assume that if the output is of the form $k \ s$, then $s = 7777777$ (seven sevens).
4. For the fourth group of test sets, worth 20 points, there are no additional assumptions.

Corresponding sample test sets are contained in `testi.in/out`, for $i \in \{1, 2, 3, 4\}$.

Sample Input		Sample Output
4	5 6 1 1 2	4 1
4 3 2 1 3	2 3 2 1 1	Busted!
2 3 1 1 1	3 1 2 1 1	4 1
3 4 1 1 1	2 4 1 1 1	3 1
4 1 1 1 1	4 1 1 1 1	
4 3 2 1 3	2 5 0 1 1	
2 3 1 1 1	5 1 0 1 1	
4 3 1 1 1	3 3 3 1 1	
4 1 1 1 1	2 1 2 1 1	
	2 3 0 1 2	
	3 1 0 1 2	